多元函数练习题及解答

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1. 多元函数的概念

1.已知$f(x,y)=x+y-xy\arctan\dfrac{x}{y}$,试求$f(tx,ty)$.

解：$f(tx,ty)=tx+ty-tx\cdot ty\arctan\dfrac{tx}{ty}=tx+ty-t^2xy\arctan\dfrac{x}{y}$.

2.试证$F(x,y)=\ln x\cdot\ln y$满足关系式

$$F(xy,uv)=F(x,u)+F(x,v)+F(y,u)+F(y,v).$$

证：左边$=(\ln xy)\cdot(\ln uv)=(\ln x+\ln y)\cdot(\ln u+\ln v)$

$$=\ln x\cdot\ln u+\ln x\cdot\ln v+\ln y\cdot\ln u+\ln y\cdot\ln v$$

$$=F(x,u)+F(x,v)+F(y,u)+F(y,v).$$

3.设$f(x+y,xy)=\dfrac{xy}{x^2+y^2}$,求$f(x,y)$.

解：令$u=x+y$，$v=xy$，我们有$f(u,v)=\dfrac{xy}{(x+y)^2-2xy}=\dfrac{v}{u^2-2v}$，即$f(x,y)=\dfrac{y}{x^2-2y}$.

4.若函数$f(x,y)$满足关系式$f(tx,ty)=t^kf(x,y)$，则称之为k次齐次函数.试证：$f(x,y)=\sqrt{x^4+y^4}-2xy$是二次齐次函数，并证明$k$次齐次函数$f(x,y)$满足$f(x,y)=x^kF\left(\dfrac{y}{x}\right)$.

证：$f(tx,ty)=\sqrt{t^4x^4+t^4y^4}-2txty=t^2\left(\sqrt{x^4+y^4}-2xy\right)=t^2f(x,y)$.

这便证明了：$f(x,y)=\sqrt{x^4+y^4}-2xy$是二次齐次函数.

观察待证式与原式的右边，不妨令$t=x$，同时保证自变量一致，可再令$tu=x$，$tv=y$，

则有$f(x,y)=f(tu,tv)=t^kf(u,v)=x^kf(u,v)=x^kf\left(1,\dfrac{y}{x}\right)$.即$f(x,y)=x^kF\left(\dfrac{y}{x}\right)$.

5.求下列函数的定义域：

(1)$z=\ln(y-2x+1)$ (2)$z=\dfrac{1}{\sqrt{x+y}}+\dfrac{1}{x-y}$ (3)$z=\sqrt{x-\sqrt{y}}$ (4)$f(x,y)=\dfrac{\sqrt{4x-y^2}}{\ln(1-x^2-y^2)}$

解(1)$y-2x+1>0$ (2)由$\begin{cases}x+y>0\\x-y\neq0\end{cases}$，有$x>-y$且$x\neq y$ (3)由$\begin{cases}x-\sqrt{y}\geqslant0\\y\geqslant0\end{cases}$，有$x\geqslant\sqrt{y}\geqslant0$.

(4)由$\begin{cases}4x-y^2\geqslant0\\1-x^2-y^2>0,\neq1\end{cases}$，有$x\geqslant\dfrac{y^2}{4}$且$0<x^2+y^2<1$.

6.求下列极限.

(1)$\lim\limits_{(x,y)\to(1,3)}\dfrac{xy}{x^2+2}$ (2)$\lim\limits_{(x,y)\to(0,0)}\dfrac{\sqrt{xy+1}-1}{xy}$ (3)$\lim\limits_{(x,y)\to(2,0)}\left(1+\dfrac{y}{x}\right)^{\frac{1}{y}}$

(4)$\lim\limits_{(x,y)\to(0,0)}\dfrac{1-\cos(x^2+y^2)}{(x^2+y^2)^2}$ (5)$\lim\limits_{(x,y)\to(0,0)}\dfrac{(y-x)x}{\sqrt{x^2+y^2}}$ (6)$\lim\limits_{(x,y)\to(0,0)}\dfrac{xy}{\sqrt{x^2+y^2}}$

解(1)原式$=\dfrac{1\cdot 3}{1^2+2}=1$. (2)原式$=\lim\limits_{t\to 0}\dfrac{\sqrt{t+1}-1}{t}=\lim\limits_{t\to 0}\dfrac{1}{\sqrt{t+1}+1}=\dfrac{1}{2}$.

(3)原式$=\lim\limits_{y\to 0}\left(1+\dfrac{y}{2}\right)^{\frac{1}{y}}=\lim\limits_{y\to 0}\left(1+\dfrac{y}{2}\right)^{\frac{2}{y}\cdot\frac{1}{2}}=\sqrt{e}$.

(4)原式$=\lim\limits_{t\to 0^+}\dfrac{1-\cos t}{t^2}=\lim\limits_{t\to 0^+}\dfrac{\sin t}{2t}=\dfrac{1}{2}$.

(5)注意到$\dfrac{|y-x|}{\sqrt{x^2+y^2}}\leqslant\dfrac{1}{\sqrt{x^2+y^2}}\cdot\sqrt{\dfrac{x^2+y^2}{2}}=\dfrac{1}{\sqrt{2}}$，则极限为0(有界·无穷小=无穷小).

(6)方法1.注意到$\dfrac{|xy|}{\sqrt{x^2+y^2}}\leqslant\dfrac{1}{\sqrt{x^2+y^2}}\cdot\dfrac{x^2+y^2}{2}=\dfrac{\sqrt{x^2+y^2}}{2}$，夹逼得极限为0.

方法2.令$\begin{cases}x=r\cos t\\ y=r\sin t\end{cases}$($t$为参数)，则$\dfrac{xy}{\sqrt{x^2+y^2}}=\dfrac{1}{2}r\sin 2t\to 0$.故极限为0.

注：受(6)的启示，方法2亦可用于(5)，留给读者解决，此处不再赘述.

7.证明下列极限不存在：(1)$\lim\limits_{(x,y)\to(0,0)}\dfrac{x+y}{x-y}$; (2)$\lim\limits_{(x,y)\to(0,0)}\dfrac{x^2y^2}{x^2y^2+(x-y)^2}$.

证(1)方法1.受题6(6)的启发，我们有$\dfrac{x+y}{x-y}=\dfrac{\cos t+\sin t}{\cos t-\sin t}$与$t$有关，非常数，证毕.

方法2.令$y=kx(k\neq 1)$，则$\dfrac{x+y}{x-y}=\dfrac{1+k}{1-k}$不是常数，故极限不存在.

(2)分析：若仍按照前6题的思维：分子、分母都是4次，故极限为1，这样显然会得到错误的结果，因x,y同为无穷小，差可是非高阶无穷小，故计算时，易出错.

解答：取$y=x$，则$\dfrac{x^2y^2}{x^2y^2+(x-y)^2}=\dfrac{x^4}{x^4}=1$，极限为1；

取$y=\dfrac{x}{2}$，则$\dfrac{x^2y^2}{x^2y^2+(x-y)^2}=\dfrac{x^2}{x^2+1}$，极限为0.故不存在.

8.函数$z=\dfrac{y^2+2x}{y^2-2x}$在何处是间断的?

解：$y^2=2x$.

拓展1.

1.设$f(x_1,x_2,\cdots,x_n)$是k次齐次函数，则$\sum\limits_{i=1}^{n}x_i\dfrac{\partial f}{\partial x_i}=kf(x_1,x_2,\cdots,x_n)$.

证：此处定义仍沿袭第4题中二元函数的定义：$f(\lambda x_1,\cdots,\lambda x_n)=\lambda^k f(x_1,\ldots,x_n)$.

对上式两边全微分，有$\dfrac{\mathrm{d}}{\mathrm{d}\lambda}f(\lambda x_1,\cdots,\lambda x_n)=\sum\limits_{i=1}^{n}\dfrac{\partial f}{\partial(\lambda x_i)}\dfrac{\mathrm{d}(\lambda x_i)}{\mathrm{d}\lambda}=\sum\limits_{i=1}^{n}x_i\dfrac{\partial f}{\partial(\lambda x_i)}$.

又$\dfrac{\mathrm{d}}{\mathrm{d}\lambda}\lambda^k f(x_1,\cdots,x_n)=k\lambda^{k-1}f(x_1,\cdots,x_n)\therefore\sum\limits_{i=1}^{n}x_i\dfrac{\partial f}{\partial(\lambda x_i)}=k\lambda^{k-1}f(x_1,\cdots,x_n)$.

取$\lambda=1$，即得$\sum\limits_{i=1}^{n}x_i\dfrac{\partial f}{\partial x_i}=kf(x_1,x_2,\cdots,x_n)$.证毕.

2.偏导数

1.求下列函数的偏导数：

(1) $z=x^3y-xy^3$ (2) $z=\sin(3x+2y^2)$ (3) $z=\sqrt{\ln(xy)}$ (4) $z=(1+xy)^y$

(5) $z=\arctan\dfrac{y}{x}+\ln\sqrt{x^2+y^2}$ (6) $s=\dfrac{u^2+v^2}{uv}$ (7) $u=x^{\frac{y}{z}}$ (8) $u=\dfrac{x+y}{y+z}$

解(1) $\dfrac{\partial z}{\partial x}=3yx^2-y^3$; $\dfrac{\partial z}{\partial y}=x^3-3xy^2$.

(2) $\dfrac{\partial z}{\partial x}=3\cos(3x+2y^2)$; $\dfrac{\partial z}{\partial y}=4y\cos(3x+2y^2)$.

(3) $\dfrac{\partial z}{\partial x}=\dfrac{1}{2\sqrt{\ln(xy)}}\cdot(\ln xy)'_x=\dfrac{1}{2\sqrt{\ln(xy)}}\cdot\dfrac{1}{x}=\dfrac{1}{2x\sqrt{\ln(xy)}}$;

由于$z=f(x,y)$对于变量是对称的，故$\dfrac{\partial z}{\partial y}=\dfrac{1}{2y\sqrt{\ln(xy)}}$(直接将$y$换成$x$).

(4)法1.由于$z=e^{y\ln(1+xy)}=e^f$，则$\dfrac{\partial z}{\partial x}=e^f\cdot\dfrac{\partial f}{\partial x}=e^f\cdot y\cdot\dfrac{y}{1+xy}=y^2(1+xy)^{y-1}$.

同时，$\dfrac{\partial z}{\partial y}=e^f\cdot\dfrac{\partial f}{\partial y}=e^f\left[\dfrac{xy}{1+xy}+\ln(1+xy)\right]=\left[\dfrac{xy}{1+xy}+\ln(1+xy)\right](1+xy)^y$.

法2.本题可以利用下一节的全微分公式来计算.

由$\ln z=y\ln(1+xy)$，有$\dfrac{\mathrm{d}z}{z}=\ln(1+xy)\mathrm{d}y+y\left(\dfrac{y\,\mathrm{d}x}{1+xy}+\dfrac{x\,\mathrm{d}y}{1+xy}\right)$.

即$\mathrm{d}z=\dfrac{y^2z}{1+xy}\mathrm{d}x+z\left[\dfrac{xy}{1+xy}+\ln(1+xy)\right]\mathrm{d}y$.

通过比较上式与$\mathrm{d}z=\dfrac{\partial z}{\partial x}\mathrm{d}x+\dfrac{\partial z}{\partial y}\mathrm{d}y$，即得$\dfrac{\partial z}{\partial x}$与$\dfrac{\partial z}{\partial y}$(结果见法1).

(5)法1.同(4)的法2.

法2. $\dfrac{\partial z}{\partial x}=\dfrac{1}{1+(y/x)^2}\cdot(-yx^{-2})+\dfrac{1}{2}\cdot\dfrac{2x}{x^2+y^2}=\dfrac{x-y}{x^2+y^2}$;

$\dfrac{\partial z}{\partial y}=\dfrac{1}{1+(y/x)^2}\cdot\dfrac{1}{x}+\dfrac{1}{2}\cdot\dfrac{2y}{x^2+y^2}=\dfrac{x+y}{x^2+y^2}$.

(6)$s=\dfrac{u}{v}+\dfrac{v}{u}$关于$u,v$是对称的，因此计算出$\dfrac{\partial s}{\partial u}=\dfrac{1}{v}-\dfrac{v}{u^2}$后，只需将$u,v$互换，

便得出$\dfrac{\partial s}{\partial v}=\dfrac{1}{u}-\dfrac{u}{v^2}$.类似题目可以同样处理，比如题1的(3).

(7)本题与(4)属于同类型，因此完全可以仿照(4)的两种方法来处理.

由$\ln u=\dfrac{y}{z}\ln x$，有$\dfrac{\mathrm{d}u}{u}=\dfrac{y}{z}\cdot\dfrac{\mathrm{d}x}{x}+\dfrac{z\,\mathrm{d}y-y\,\mathrm{d}z}{z^2}\ln x$.即$\mathrm{d}u=\dfrac{yu}{xz}\mathrm{d}x+\dfrac{u\ln x}{z}\mathrm{d}y-\dfrac{yu\ln x}{z^2}\mathrm{d}z$.

故$\dfrac{\partial u}{\partial x}=\dfrac{y}{z}x^{\frac{y}{z}-1}$; $\dfrac{\partial u}{\partial y}=\dfrac{1}{z}x^{\frac{y}{z}}\ln x$; $\dfrac{\partial u}{\partial z}=-\dfrac{y}{z^2}x^{\frac{y}{z}}\ln x$.

(8) $\dfrac{\partial u}{\partial x}=\dfrac{1}{y+z}$; $\dfrac{\partial u}{\partial y}=\dfrac{z-x}{(y+z)^2}$; $\dfrac{\partial u}{\partial z}=-\dfrac{x+y}{(y+z)^2}$.

注：有具体表达式的多元函数求偏导除直接偏导外，可以仿照题(4)求全微分处理，这样做的的好处是可以

一次性的得出所有偏导数，不失为一种简便的方法.另外需要强调的是具有轮换对称性的函数，求完偏导后只需将自变量互换，便得出所有的偏导数.在接下来的解答中，前文提及的内容不再赘述，并建议读者掌握本题中所提及的两种计算偏导数的方法，并会熟练运用.

2.$f(x,y)=x+(y-1)\arcsin\sqrt{\dfrac{x}{y}}$，求$f_x(x,1)$.

解：$f_x(x,y)=1+(y-1)\dfrac{\partial}{\partial x}\left(\arcsin\sqrt{\dfrac{x}{y}}\right)$，故$f_x(x,1)=1+(1-1)\cdot g(x,1)=1$.

3.设$u=\ln(1+x^2+y^2+z^2)$，求$u_x+u_y+u_z$.

解：$u_x=\dfrac{2x}{1+x^2+y^2+z^2}$，故$u_x+u_y+u_z=\dfrac{2(x+y+z)}{1+x^2+y^2+z^2}$(轮换对称性,上文已提及).

4.曲线$\begin{cases}z=\dfrac{x^2+y^2}{4}\\ y=4\end{cases}$,在点$(2,4,5)$处的切线对于$x$轴的倾角是多少？

解：$z_x(2,4)=\left.\dfrac{x}{2}\right|_{x=2}=1$，倾角为$\dfrac{\pi}{4}$.

5.求曲线$\begin{cases}z=\sqrt{1+x^2+y^2}\\ x=1\end{cases}$在点$\left(1,1,\sqrt{3}\right)$的切线与$y$轴的正向所成的角度.

解：$z_y(1,1)=\left.\dfrac{y}{\sqrt{1+x^2+y^2}}\right|_{(1,1)}=\dfrac{1}{\sqrt{3}}$，角度为$\dfrac{\pi}{6}$.

6.设$z=xy+xe^{\frac{y}{x}}$，验证$x\dfrac{\partial z}{\partial x}+y\dfrac{\partial z}{\partial y}=xy+z$.

解：$\dfrac{\partial z}{\partial x}=y+e^{\frac{y}{x}}+xe^{\frac{y}{x}}\cdot\dfrac{-y}{x^2}=y+\left(1-\dfrac{y}{x}\right)e^{\frac{y}{x}}$；$\dfrac{\partial z}{\partial y}=x+xe^{\frac{y}{x}}\cdot\dfrac{1}{x}=x+e^{\frac{y}{x}}$.

故$x\dfrac{\partial z}{\partial x}+y\dfrac{\partial z}{\partial y}=2xy+xe^{\frac{y}{x}}=xy+z$.

7.设$z=e^{-\left(\frac{1}{x}+\frac{1}{y}\right)}$，验证$x^2\dfrac{\partial z}{\partial x}+y^2\dfrac{\partial z}{\partial y}=2z$.

解：$\dfrac{\partial z}{\partial x}=e^{-\left(\frac{1}{x}+\frac{1}{y}\right)}\cdot\dfrac{1}{x^2}$，则$x^2\dfrac{\partial z}{\partial x}=z$，同理$y^2\dfrac{\partial z}{\partial y}=z$，故$x^2\dfrac{\partial z}{\partial x}+y^2\dfrac{\partial z}{\partial y}=2z$.

8.求下列函数的二阶偏导数$\dfrac{\partial^2 z}{\partial x^2},\dfrac{\partial^2 z}{\partial y^2},\dfrac{\partial^2 z}{\partial x\partial y}$：$(1)z=x^4+y^4-4x^2y^2\quad(2)z=\arctan\dfrac{y}{x}$.

解(1)$\dfrac{\partial z}{\partial x}=4x^3-8y^2x;\dfrac{\partial z}{\partial y}=4y^3-8x^2y;\dfrac{\partial^2 z}{\partial x^2}=12x^2-8y^2;\dfrac{\partial^2 z}{\partial y^2}=12y^2-8x^2;\dfrac{\partial^2 z}{\partial x\partial y}=-16xy.$

9.设$z=(x^2+y)\sin(xy)$，求$\dfrac{\partial^2 z}{\partial x^2}$.

解：$\dfrac{\partial z}{\partial x}=2x\sin xy+(x^2+y)\cos xy\cdot y;$

$\dfrac{\partial^2 z}{\partial x^2}=2\sin xy+2xy\cos xy+2xy\cos xy-y^2(x^2+y)\sin xy=(2-x^2y^2-y^3)\sin xy+4xy\cos xy.$

10.验证：

(1) $z=\ln(e^x+e^y)$ 满足 $\dfrac{\partial^2 z}{\partial x^2}\cdot\dfrac{\partial^2 z}{\partial y^2}-\left(\dfrac{\partial^2 z}{\partial x\partial y}\right)^2=0.$

(2) $u=r=\sqrt{x^2+y^2+z^2}$ 满足 $\dfrac{\partial^2 u}{\partial x^2}+\dfrac{\partial^2 u}{\partial y^2}+\dfrac{\partial^2 u}{\partial z^2}=\dfrac{2}{r}.$

解 (1) $\dfrac{\partial z}{\partial x}=\dfrac{e^x}{e^x+e^y};\ \dfrac{\partial^2 z}{\partial x^2}=\dfrac{\partial^2 z}{\partial y^2}=\dfrac{e^{x+y}}{(e^x+e^y)^2};\ \dfrac{\partial^2 z}{\partial x\partial y}=-\dfrac{e^{x+y}}{(e^x+e^y)^2}.$

故左边 $=\dfrac{e^{2(x+y)}}{(e^x+e^y)^4}-\dfrac{e^{2(x+y)}}{(e^x+e^y)^4}=0.$

(2) $\dfrac{\partial u}{\partial x}=\dfrac{x}{\sqrt{x^2+y^2+z^2}}=\dfrac{x}{r};\ \dfrac{\partial^2 u}{\partial x^2}=\dfrac{1\cdot r-x\dfrac{\partial r}{\partial x}}{r^2}=\dfrac{r^2-x^2}{r^3}.$

于是，左边 $=\dfrac{3r^2-(x^2+y^2+z^2)}{r^3}=\dfrac{2r^2}{r^3}=\dfrac{2}{r}.$

11.设 $z=f(x,y)$ 满足 $f_{yy}(x,y)=2$ 且 $f(x,0)=1$，$f_y(x,0)=x$.求 $f(x,y)$.

解：$f_y(x,y)=\int f_{yy}(x,y)\mathrm{d}y=2y+u(x)$.令 $y=0$，有 $f_y(x,0)=u(x)=x.$

$\therefore f_y(x,y)=x+2y\Rightarrow f(x,y)=\int f_y(x,y)\mathrm{d}y=xy+y^2+v(x).$

令 $y=0$，有 $f(x,0)=v(x)=1$.故 $f(x,y)=xy+y^2+1.$

3. 全微分

1.求下列函数的全微分：

(1) $z=xy+\dfrac{x}{y}$ (2) $z=\dfrac{x^2-y^2}{x^2+y^2}$ (3) $u=\arctan\dfrac{xy}{z^2}$ (4) $u=\left(xy+\dfrac{x}{y}\right)^z$

解 (1) $\mathrm{d}z=x\,\mathrm{d}y+y\,\mathrm{d}x+\dfrac{y\,\mathrm{d}x-x\,\mathrm{d}y}{y^2}=(y+y^{-1})\mathrm{d}x+\left(x-\dfrac{x}{y^2}\right)\mathrm{d}y.$

(2) $\mathrm{d}z=\dfrac{(x^2+y^2)(2x\,\mathrm{d}x-2y\,\mathrm{d}y)-(x^2-y^2)(2x\,\mathrm{d}x+2y\,\mathrm{d}y)}{(x^2+y^2)^2}=\dfrac{4xy(y\,\mathrm{d}x-x\,\mathrm{d}y)}{(x^2+y^2)^2}.$

(3) $\mathrm{d}u=\dfrac{1}{1+(xy/z^2)^2}\cdot\dfrac{z^2\mathrm{d}(xy)-xy\,\mathrm{d}(z^2)}{z^4}=\dfrac{yz^2\mathrm{d}x+xz^2\mathrm{d}y-2xyz\,\mathrm{d}z}{x^2y^2+z^4}.$

(4) 由 $\ln u=z\ln\left(xy+\dfrac{x}{y}\right)$，有 $\dfrac{\mathrm{d}u}{u}=\ln\left(xy+\dfrac{x}{y}\right)\mathrm{d}z+z\left[\dfrac{\mathrm{d}x}{x}+\left(\dfrac{2y}{y^2+1}-\dfrac{1}{y}\right)\mathrm{d}y\right].$

于是，$\mathrm{d}u=\left(xy+\dfrac{x}{y}\right)^{y-1}\left[\left(y+\dfrac{1}{y}\right)z\,\mathrm{d}x+\left(1-\dfrac{1}{y^2}\right)xz\,\mathrm{d}y+\left(xy+\dfrac{x}{y}\right)\ln\left(xy+\dfrac{x}{y}\right)\mathrm{d}z\right].$

2.求 $z=\ln(1+x^2+y^2)$ 当 $x=1,y=2$ 时的全微分.

解：$\mathrm{d}z|_{(1,2)}=\left.\dfrac{2x\,\mathrm{d}x+2y\,\mathrm{d}y}{1+x^2+y^2}\right|_{(1,2)}=\dfrac{\mathrm{d}x+2\,\mathrm{d}y}{3}.$

3.求$z=\dfrac{xy}{x^2-y^2}$当$x=2,y=1,\Delta x=0.01,\Delta y=0.03$时的全微分.

解：$\mathrm{d}z=\dfrac{2xy(x\,\mathrm{d}x-y\,\mathrm{d}y)-(x^2-y^2)(y\,\mathrm{d}x+x\,\mathrm{d}y)}{(x^2-y^2)^2}$.

将条件代入，有$\mathrm{d}z=\dfrac{-2\cdot2(2\cdot0.01-0.03)+(2^2-1^2)(0.01+2\cdot0.03)}{(2^2-1^2)^2}=\dfrac{1}{36}$.

4.设R是电阻R_1,R_2并联后的总电阻.

(1)证明$\mathrm{d}R=\left(\dfrac{R}{R_1}\right)^2\mathrm{d}R_1+\left(\dfrac{R}{R_2}\right)^2\mathrm{d}R_2$.

(2)现假设$R_1=100\Omega$，$R_2=400\Omega$，则总电阻R对哪个电阻的误差更敏感？

解(1)由$\dfrac{1}{R}=\dfrac{1}{R_1}+\dfrac{1}{R_2}$，有$\dfrac{\mathrm{d}R}{R^2}=\dfrac{\mathrm{d}R_1}{R_1^2}+\dfrac{\mathrm{d}R_2}{R_2^2}$.即$\mathrm{d}R=\left(\dfrac{R}{R_1}\right)^2\mathrm{d}R_1+\left(\dfrac{R}{R_2}\right)^2\mathrm{d}R_2$.

(2)由于$\left(\dfrac{R}{R_1}\right)^2>\left(\dfrac{R}{R_2}\right)^2$，故当$R_1,R_2$分别增加单位值时，$R$对$R_1$的误差更敏感.

5.求函数$f(x,y,z)=xy+yz+zx$在下列各点的线性化函数$L(x,y,z)$.

(1)$(1,1,1)$　　(2)$(1,0,0)$　　(3)$(1,2,3)$.

解：$f_x=y+z$，$f_y=x+z$，$f_z=x+y$.

(1)$f_x=f_y=f_z=2$，故$L(x,y,z)=3+2(x-1)+2(y-1)+2(z-1)=2x+2y+2z-3$.

(2)$f_x=0,f_y=f_z=1$，故$L(x,y,z)=y+z$.

(3)$f_x=5,f_y=4,f_z=3$，故$L(x,y,z)=5x+4y+3z-11$.

6.设函数$f(x,y)=\begin{cases}\dfrac{xy}{\sqrt{x^2+y^2}},x^2+y^2\neq0\\0,\qquad x^2+y^2=0\end{cases}$.

(1)函数$f(x,y)$在点$(0,0)$是否连续？

(2)偏导数$f_x(0,0),f_y(0,0)$是否存在？

(3)函数$f(x,y)$在点$(0,0)$处是否可微分？

解(1)由习题7.1的题6(6)，有$\lim\limits_{(x,y)\to(0,0)}f(x,y)=f(0,0)=0$.故连续.

(2)本函数是本节所举的例，故存在，且均为0.

(3)同(2)，不可微.

7.讨论函数$f(x,y)=\sqrt{|xy|}$在点$(0,0)$处的连续性，可导性，可微性.

解：$\lim\limits_{(x,y)\to(0,0)}f(x,y)=f(0,0)=0$，故连续.

由$\lim\limits_{x\to0}\dfrac{0-0}{x-0}=0$，知$f_x(0,0)=0$.同理$f_y(0,0)=0$，故可导.

因$\lim\limits_{(x,y)\to(0,0)}\dfrac{\sqrt{|xy|}-0-0}{\sqrt{x^2+y^2}}$不存在，知不可微.

8.求函数$f(x,y)=\begin{cases}(x^2+y^2)\sin[(x^2+y^2)^{-1}],x^2+y^2\neq0\\0,\qquad\qquad\qquad x^2+y^2=0\end{cases}$的偏导数，并讨论函数在点$(0,0)$处的偏导数的

连续性及函数可微性.

解：由$\lim\limits_{h\to 0}\dfrac{f(h,0)-f(0,0)}{h}=\lim\limits_{h\to 0}\dfrac{h^2\sin\frac{1}{h^2}}{h}=\lim\limits_{h\to 0}h\sin\dfrac{1}{h^2}=0$，有$f_x(0,0)$.

当$x^2+y^2\neq 0$时，$f_x=2x\sin\dfrac{1}{x^2+y^2}+(x^2+y^2)\cos\dfrac{1}{x^2+y^2}\cdot\dfrac{-2x}{(x^2+y^2)^2}$

$$=2x\sin\frac{1}{x^2+y^2}-\frac{2x}{x^2+y^2}\cos\frac{1}{x^2+y^2}.$$

故$f_x(x,y)=\begin{cases}2x\sin\dfrac{1}{x^2+y^2}-\dfrac{2x}{x^2+y^2}\cos\dfrac{1}{x^2+y^2}, & x^2+y^2\neq 0\\ 0, & x^2+y^2=0\end{cases}$.

同理$f_y(x,y)=\begin{cases}2y\sin\dfrac{1}{x^2+y^2}-\dfrac{2y}{x^2+y^2}\cos\dfrac{1}{x^2+y^2}, & x^2+y^2\neq 0\\ 0, & x^2+y^2=0\end{cases}$.

由$\lim\limits_{x\to 0}f_x(x,0)$不存在，知$f_x(x,y)$在$(0,0)$不连续，同理$f_y(x,y)$在$(0,0)$不连续.

注意到$\lim\limits_{(x,y)\to 0}\dfrac{f(x,y)-f(0,0)}{\sqrt{x^2+y^2}}=\lim\limits_{(x,y)\to(0,0)}\sqrt{x^2+y^2}\sin\dfrac{1}{x^2+y^2}=0$，故可微.

4. 多元复合函数的求导法则

1.设$z=u^2+v^2$，而$u=x+y,v=x-y$，求$\dfrac{\partial z}{\partial x},\dfrac{\partial z}{\partial y}$.

解：$\dfrac{\partial z}{\partial x}=\dfrac{\partial z}{\partial u}\dfrac{\partial u}{\partial x}+\dfrac{\partial z}{\partial v}\dfrac{\partial v}{\partial x}=2(u+v)=4x$;由于$z$关于$x,y$对称，故$\dfrac{\partial z}{\partial y}=4y$.

2.设$z=\arctan\dfrac{y}{x}$，$x=u+v$，$y=uv$，求$\dfrac{\partial z}{\partial u},\dfrac{\partial z}{\partial v}$.

解：$\dfrac{\partial z}{\partial u}=\dfrac{\partial z}{\partial x}\dfrac{\partial x}{\partial u}+\dfrac{\partial z}{\partial y}\dfrac{\partial y}{\partial u}=\dfrac{-y}{x^2+y^2}+\dfrac{x}{x^2+y^2}v=\dfrac{v^2}{(u+v)^2+u^2v^2}$;

$\dfrac{\partial z}{\partial v}=\dfrac{\partial z}{\partial x}\dfrac{\partial x}{\partial v}+\dfrac{\partial z}{\partial y}\dfrac{\partial y}{\partial v}=\dfrac{-y}{x^2+y^2}+\dfrac{x}{x^2+y^2}u=\dfrac{u^2}{(u+v)^2+u^2v^2}$.

3.设$z=\arcsin(x-y)$，而$x=3t,y=4t^3$，求$\dfrac{\mathrm{d}z}{\mathrm{d}t}$.

解：$z=\arcsin(3t-4t^3)$，故$\dfrac{\mathrm{d}z}{\mathrm{d}t}=\dfrac{3-12t^2}{\sqrt{1-(3t-4t^3)^2}}$.

4.设$y=u^v$，而$u=u(x),v=v(x)$都是可导函数，求$\dfrac{\mathrm{d}y}{\mathrm{d}x}$.

解：$\dfrac{\mathrm{d}y}{\mathrm{d}x}=\dfrac{\partial y}{\partial u}\dfrac{\mathrm{d}u}{\mathrm{d}x}+\dfrac{\partial y}{\partial v}\dfrac{\mathrm{d}v}{\mathrm{d}x}=vu^{v-1}u'+u^v\ln v\cdot v'$.

6.设$z=\arctan\dfrac{x}{y}$，而$x=u+v,y=u-v$，验证：$\dfrac{\partial z}{\partial u}+\dfrac{\partial z}{\partial v}=\dfrac{u-v}{u^2+v^2}$.

解：$\dfrac{\partial z}{\partial u}=\dfrac{\partial z}{\partial x}\dfrac{\partial x}{\partial u}+\dfrac{\partial z}{\partial y}\dfrac{\partial y}{\partial u}=\dfrac{y-x}{x^2+y^2}$.同理，$\dfrac{\partial z}{\partial v}=\dfrac{x+y}{x^2+y^2}$.

故 $\dfrac{\partial z}{\partial u}+\dfrac{\partial z}{\partial v}=\dfrac{2y}{x^2+y^2}=\dfrac{u-v}{u^2+v^2}.$

7.设 $z=xy+xF(u)$，而 $u=\dfrac{y}{x}$，$F(u)$ 是可导函数，证明：$x\dfrac{\partial z}{\partial x}+y\dfrac{\partial z}{\partial y}=z+xy.$

证：$\dfrac{\partial z}{\partial x}=y+F(u)+xF'(u)\dfrac{-y}{x^2}=y+F(u)-\dfrac{y}{x}F'(u);$

$\dfrac{\partial z}{\partial y}=x+xF'(u)\cdot\dfrac{1}{x}=x+F'(u).$

故 $x\dfrac{\partial z}{\partial x}+y\dfrac{\partial z}{\partial y}=xy+xF(u)-yF'(u)+xy+yF'(u)=z+xy.$

8.设 $z=x^3f\left(\dfrac{y}{x^2}\right)$，$f$ 是可导函数，证明：$x\dfrac{\partial z}{\partial x}+2y\dfrac{\partial z}{\partial y}=3z.$

证：$\dfrac{\partial z}{\partial x}=3x^2f\left(\dfrac{y}{x^2}\right)+x^3f'\left(\dfrac{y}{x^2}\right)\cdot\dfrac{-2y}{x^3};\ \dfrac{\partial z}{\partial y}=x^3f'\left(\dfrac{y}{x^2}\right)\dfrac{1}{x^2}=xf'\left(\dfrac{y}{x^2}\right).$

故 $x\dfrac{\partial z}{\partial x}+2y\dfrac{\partial z}{\partial y}=3x^3f\left(\dfrac{y}{x^2}\right)-2xyf'\left(\dfrac{y}{x^2}\right)+2xyf'\left(\dfrac{y}{x^2}\right)=3z.$

9.设 $z=f(x+\varphi(y))$，其中 f,φ 具有二阶导数，求证：$\dfrac{\partial z}{\partial x}\cdot\dfrac{\partial^2 z}{\partial x\partial y}=\dfrac{\partial z}{\partial y}\cdot\dfrac{\partial^2 z}{\partial x^2}.$

证：$\dfrac{\partial z}{\partial x}=f';\ \dfrac{\partial z}{\partial y}=f'\varphi';\ \dfrac{\partial^2 z}{\partial x\partial y}=f''\varphi';\ \dfrac{\partial^2 z}{\partial x^2}=f''.$

故 $\dfrac{\partial z}{\partial x}\cdot\dfrac{\partial^2 z}{\partial x\partial y}-\dfrac{\partial z}{\partial y}\cdot\dfrac{\partial^2 z}{\partial x^2}=f'f''\varphi'-f'\varphi'f''=0.$

10.设 $z=f(x^2+y^2)$，求 $\dfrac{\partial^2 z}{\partial x^2},\dfrac{\partial^2 z}{\partial y^2},\dfrac{\partial^2 z}{\partial x\partial y}.$

解：$\dfrac{\partial z}{\partial x}=2xf'(x^2+y^2);\ \dfrac{\partial^2 z}{\partial x^2}=2f'(x^2+y^2)+4x^2f''(x^2+y^2).$

同理 $\dfrac{\partial^2 z}{\partial y^2}=2f'(x^2+y^2)+4y^2f''(x^2+y^2).$同时，$\dfrac{\partial^2 z}{\partial x\partial y}=4xyf''(x^2+y^2).$

11.设 $z=f(x,g(x,y))$，$y=h(x)$，其中 f 有连续偏导数，h 是可导函数，求 $\dfrac{\mathrm{d}z}{\mathrm{d}x}.$

解：$\dfrac{\mathrm{d}z}{\mathrm{d}x}=f_1+f_2\dfrac{\partial g}{\partial x}=f_1+f_2\left(g_1+g_2\dfrac{\mathrm{d}h}{\mathrm{d}x}\right)=(f_1+f_2g_1)+f_2g_2\dfrac{\mathrm{d}h}{\mathrm{d}x}.$

12.设 $g(x,t)=\displaystyle\int_{x-at}^{x+at}f(u)\mathrm{d}u$，其中 $f(u)$ 在 $(-\infty,\infty)$ 可导，求 $\dfrac{\partial^2 g}{\partial x^2},\dfrac{\partial^2 g}{\partial t^2}.$

解：$\dfrac{\partial g}{\partial x}=f(x+at)-f(x-at);\ \dfrac{\partial^2 g}{\partial x^2}=f'(x+at)-f'(x-at).$

$\dfrac{\partial g}{\partial t}=af(x+at)+af(x-at);\ \dfrac{\partial^2 g}{\partial t^2}=a^2[f'(x+at)-f'(x-at)].$

13.设 f 有二阶连续偏导数，求 $\dfrac{\partial^2 z}{\partial x^2},\dfrac{\partial^2 z}{\partial y^2},\dfrac{\partial^2 z}{\partial x\partial y}$，其中

(1)设 $z=f(xy,y);$

(2)设 $z=f(xy^2,x^2y);$

(3)设 $z=f(\sin x,\cos y,e^{x+y}).$

解 (1) $\dfrac{\partial z}{\partial x}=yf_1;\ \dfrac{\partial z}{\partial y}=xf_1+f_2;\ \dfrac{\partial^2 z}{\partial x^2}=yf_{11};\ \dfrac{\partial^2 z}{\partial x\partial y}=f_1+y(xf_{11}+f_{12}).$

$\dfrac{\partial^2 z}{\partial y^2}=x(xf_{11}+f_{12})+(xf_{21}+f_{22})=x^2f_{11}+2xf_{12}+f_{22}.$

(2) $\dfrac{\partial z}{\partial x}=y^2f_1+2xyf_2;\ \dfrac{\partial z}{\partial y}=2xyf_1+x^2f_2.$

$\dfrac{\partial^2 z}{\partial x^2}=y^2(y^2f_{11}+2xyf_{12})+2yf_2+2xy(y^2f_{21}+2xyf_{22})=y^4f_{11}+4xy^3f_{12}+4x^2y^2f_{22}+2yf_2.$

$\dfrac{\partial^2 z}{\partial x\partial y}=2yf_1+y^2(2xyf_{11}+x^2f_{12})+2xf_2+2xy(2xyf_{21}+x^2f_{22})$

$=2yf_1+2xf_2+2xy^3f_{11}+5x^2y^2f_{12}+2x^3yf_{22}.$

$\dfrac{\partial^2 z}{\partial y^2}=2xf_1+2xy(2xyf_{11}+x^2f_{12})+x^2(2xyf_{21}+x^2f_{22})=2xf_1+4x^2y^2f_{11}+4x^3yf_{12}+x^4f_{22}.$

(3) $\dfrac{\partial z}{\partial x}=f_1\cos x+f_3e^{x+y};\ \dfrac{\partial z}{\partial y}=-f_2\sin y+f_3e^{x+y}.$

$\dfrac{\partial^2 z}{\partial x^2}=-f_1\sin x+(f_{11}\cos x+f_{13}e^{x+y})\cos x+f_3e^{x+y}+(f_{31}\cos x+f_{33}e^{x+y})e^{x+y}$

$=f_3e^{x+y}-f_1\sin x+f_{11}\cos^2 x+2f_{13}e^{x+y}\cos x+f_{33}e^{2(x+y)}.$

$\dfrac{\partial^2 z}{\partial x\partial y}=(-f_{12}\sin y+f_{13}e^{x+y})\cos x+f_3e^{x+y}+(-f_{32}\sin y+f_{33}e^{x+y})e^{x+y}$

$=f_3e^{x+t}-f_{12}\sin y\cos x+f_{13}e^{x+y}\cos x-f_{32}e^{x+y}\sin y+f_{33}e^{2(x+y)}.$

$\dfrac{\partial^2 z}{\partial y^2}=-(-f_{22}\sin y+f_{23}e^{x+y})\sin y-f_2\cos y+f_3e^{x+y}+(-f_{32}\sin y+f_{33}e^{x+y})e^{x+y}$

$=f_3e^{x+y}-f_2\cos y+f_{22}\sin^2 y-2f_{23}e^{x+y}\sin y+f_{33}e^{2(x+y)}.$

14.设 $u=f(x,y)$ 所有的二阶偏导数连续，而 $x=\dfrac{s-\sqrt{3}t}{2},y=\dfrac{\sqrt{3}s+t}{2}$.证明：

$\left(\dfrac{\partial u}{\partial x}\right)^2+\left(\dfrac{\partial u}{\partial y}\right)^2=\left(\dfrac{\partial u}{\partial s}\right)^2+\left(\dfrac{\partial u}{\partial t}\right)^2$ 及 $\dfrac{\partial^2 u}{\partial x^2}+\dfrac{\partial^2 u}{\partial y^2}=\dfrac{\partial^2 u}{\partial s^2}+\dfrac{\partial^2 u}{\partial t^2}.$

证：$\dfrac{\partial u}{\partial s}=\dfrac{1}{2}f_1+\dfrac{\sqrt{3}}{2}f_2;\ \dfrac{\partial u}{\partial t}=-\dfrac{\sqrt{3}}{2}f_1+\dfrac{1}{2}f_2.$

$\dfrac{\partial^2 u}{\partial s^2}=\dfrac{1}{4}f_{11}+\dfrac{\sqrt{3}}{2}f_{12}+\dfrac{3}{4}f_{22};\ \dfrac{\partial^2 u}{\partial t^2}=\dfrac{3}{4}f_{11}-\dfrac{\sqrt{3}}{2}f_{12}+\dfrac{1}{4}f_{22}.$

故 $\left(\dfrac{\partial u}{\partial x}\right)^2+\left(\dfrac{\partial u}{\partial y}\right)^2=f_1^2+f_2^2;\ \dfrac{\partial^2 u}{\partial s^2}+\dfrac{\partial^2 u}{\partial t^2}=f_{11}+f_{22}.$

15.若函数 $f(x,y)$ 满足 $f(x,2x)=x^2+3x$，$f_x(x,2x)=6x+1$，求 $f_y(x,2x)$.

解：由 $\dfrac{\mathrm{d}f}{\mathrm{d}x}=f_x+2f_y$，有 $f_y(x,2x)=\dfrac{f'(x,2x)-f_x(x,2x)}{2}=1-2x.$

16.设函数 $z=f(x,y)$ 在点 $(1,1)$ 处可微，且 $f(1,1)=1$，$\left.\dfrac{\partial f}{\partial x}\right|_{(1,1)}=2$，$\left.\dfrac{\partial f}{\partial y}\right|_{(1,1)}=3$.再设函数 $\varphi(x)$ $=f(x,f(x,x))$，求 $\dfrac{\mathrm{d}}{\mathrm{d}x}\varphi^3(x)|_{x=1}$.

解：$\dfrac{\mathrm{d}\varphi^3}{\mathrm{d}x}=3\varphi^2\dfrac{\mathrm{d}\varphi}{\mathrm{d}x}=3\varphi^2[f_x+f_y(f_x+f_y)]=3[2+2(2+3)]=36.$

17.设函数$z=f(\varphi(x)-y,x+\psi(y))$，其中$f$具有二阶连续偏导数，$\varphi,\psi$可微.

求偏导数$\dfrac{\partial z}{\partial x},\dfrac{\partial^2 z}{\partial x^2},\dfrac{\partial^2 z}{\partial x\partial y}$.

解：$\dfrac{\partial z}{\partial x}=\varphi' f_1+f_2;$

$\dfrac{\partial^2 z}{\partial x^2}=\varphi'' f_1+\varphi'(\varphi' f_{11}+f_{12})+(\varphi' f_{21}+f_{22})=\varphi'' f_1+\varphi'^2 f_{11}+2\varphi' f_{12}+f_{22};$

$\dfrac{\partial^2 z}{\partial x\partial y}=\varphi'(-f_{11}+f_{12}\psi')+(-f_{21}+f_{22}\psi')=\psi' f_{22}+(\varphi'\psi'-1)f_{12}-\varphi' f_{11}.$

18.设函数φ,ψ都具有连续的一阶与二阶偏导数，且$z=\dfrac{1}{2}[\varphi(y+ax)+\varphi(y-ax)]+\dfrac{1}{2a}\displaystyle\int_{y-ax}^{y+ax}\psi(t)\mathrm{d}t.$

证明：$\dfrac{\partial^2 z}{\partial x^2}-a^2\dfrac{\partial^2 z}{\partial y^2}=0.$

证：$\dfrac{\partial z}{\partial x}=\dfrac{a}{2}[\varphi'(y+ax)-\varphi'(y-ax)]+\dfrac{1}{2}[\psi(y+ax)+\psi(y-ax)].$

$\dfrac{\partial^2 z}{\partial x^2}=\dfrac{a^2}{2}[\varphi''(y+ax)+\varphi''(y-ax)]+\dfrac{a}{2}[\psi'(y+ax)-\psi'(y-ax)].$

$\dfrac{\partial z}{\partial y}=\dfrac{1}{2}[\varphi'(y+ax)+\varphi'(y-ax)]+\dfrac{1}{2a}[\psi(y+ax)-\psi(y-ax)].$

$\dfrac{\partial^2 z}{\partial y^2}=\dfrac{1}{2}[\varphi''(y+ax)+\varphi''(y-ax)]+\dfrac{1}{2a}[\psi'(y+ax)-\psi'(y-ax)]$.代入即证毕.

5.隐函数的求导公式

1.设$\sin y+e^x-xy^2=0$，求$\dfrac{\mathrm{d}y}{\mathrm{d}x}$.

解：两边同时微分，有$\cos y\,\mathrm{d}y+e^x\mathrm{d}x-y^2\mathrm{d}x-2xy\,\mathrm{d}y=0.$

整理可得：$\dfrac{\mathrm{d}y}{\mathrm{d}x}=\dfrac{y^2-e^x}{\cos y-2xy}.$

注：本题亦可按照下面第3题，利用隐函数求导来解决.

2.设$\ln\sqrt{x^2+y^2}=\arctan\dfrac{y}{x}$，求$\dfrac{\mathrm{d}y}{\mathrm{d}x}$.

解：两边同时微分，有$\dfrac{x\,\mathrm{d}x+y\,\mathrm{d}y}{x^2+y^2}=\dfrac{(x\,\mathrm{d}y-y\,\mathrm{d}x)/x^2}{1+(y/x)^2}.$

即$x\,\mathrm{d}x+y\,\mathrm{d}y=x\,\mathrm{d}y-y\,\mathrm{d}x$.整理可得：$\dfrac{\mathrm{d}y}{\mathrm{d}x}=\dfrac{x-y}{x+y}.$

3.设$x+2y+z-2\sqrt{xyz}=0$，求$\dfrac{\partial z}{\partial x},\dfrac{\partial z}{\partial y}$.

解：设$F(x,y,z)=x+2y+z-2\sqrt{xyz}$，则

$$F_x=1-\frac{\sqrt{xyz}}{x},\quad F_y=2-\frac{\sqrt{xyz}}{y},\quad F_z=1-\frac{\sqrt{xyz}}{z}.$$

故 $\dfrac{\partial z}{\partial x}=-\dfrac{z}{x}\cdot\dfrac{x-\sqrt{xyz}}{z-\sqrt{xyz}}$，$\dfrac{\partial z}{\partial y}=-\dfrac{z}{y}\cdot\dfrac{2y-\sqrt{xyz}}{z-\sqrt{xyz}}$.

注：如果将原式代入上述结果，会有不同形式的答案，此处则不给出.

4.设 $\dfrac{x}{z}=\ln\dfrac{z}{y}$，求 $\dfrac{\partial z}{\partial x},\dfrac{\partial z}{\partial y}$.

解：设 $F(x,y,z)=\dfrac{x}{z}-\ln\dfrac{z}{y}$，则 $F_x=\dfrac{1}{z}$，$F_y=\dfrac{1}{y}$，$F_z=\dfrac{z-x}{z^2}$.

故 $\dfrac{\partial z}{\partial x}=-\dfrac{F_x}{F_z}=\dfrac{z}{x-z}$，$\dfrac{\partial z}{\partial y}=-\dfrac{F_y}{F_z}=\dfrac{z^2}{y(x-z)}$.

5.设 $2\sin(x+2y-3z)=x+2y-3z$，证明：$\dfrac{\partial z}{\partial x}+\dfrac{\partial z}{\partial y}=1$.

证：设 $u=x+2y-3z$，且 $F(u)=2\sin u-u=0$.

于是，$F_x=2\cos u\dfrac{\partial u}{\partial x}-\dfrac{\partial u}{\partial x}=2\cos u-1$，$F_y=4\cos u-2$，$F_z=-6\cos u+3$.

故 $\dfrac{\partial z}{\partial x}+\dfrac{\partial z}{\partial y}=-\dfrac{F_x+F_y}{F_z}=-\dfrac{2\cos u-1+4\cos u-2}{-6\cos u+3}=\dfrac{6\cos u-3}{6\cos u-3}=1$.

6.设 $x=x(y,z),y=y(x,z),z=z(x,y)$ 都是由方程 $F(x,y,z)=0$ 所确定的具有连续偏导数的函数.

证明：$\dfrac{\partial x}{\partial y}\cdot\dfrac{\partial y}{\partial z}\cdot\dfrac{\partial z}{\partial x}=-1$.

证：注意到 $\dfrac{\partial x}{\partial y}=-\dfrac{F_y}{F_x}$，$\dfrac{\partial y}{\partial z}=-\dfrac{F_z}{F_y}$，$\dfrac{\partial z}{\partial x}=-\dfrac{F_x}{F_z}$，故 $\dfrac{\partial x}{\partial y}\cdot\dfrac{\partial y}{\partial z}\cdot\dfrac{\partial z}{\partial x}=-1$.

7.设 $F(u,v)$ 有连续偏导数，且 $F(x^2-y^2,y^2-z^2)=0$.求 $\dfrac{\partial z}{\partial x},\dfrac{\partial z}{\partial y}$.

解：$F_x=2xF_1$，$F_y=-2yF_1+2yF_2$，$F_z=-2zF_2$.

因此，$\dfrac{\partial z}{\partial x}=-\dfrac{F_x}{F_z}=\dfrac{xF_1}{zF_2}$，$\dfrac{\partial z}{\partial y}=-\dfrac{F_y}{F_z}=\dfrac{y(F_2-F_1)}{zF_2}$.

8.设 $z=z(x,y)$ 由方程 $x-az=\varphi(y-bz)$ 确定，其中 $\varphi(u)$ 有连续导数，证明 $a\dfrac{\partial z}{\partial x}+b\dfrac{\partial z}{\partial y}=1$.

证：设 $F(x,y,z)=x-az-\varphi(y-bz)$，则 $F_x=1$，$F_y=-\varphi'(y-bz)$，$F_z=b\varphi'(y-bz)-a$.

故 $a\dfrac{\partial z}{\partial x}+b\dfrac{\partial z}{\partial y}=-\dfrac{aF_x+bF_y}{F_z}=\dfrac{-a+b\varphi'(y-bz)}{b\varphi'(y-bz)-a}=1$.

9.设 $F(u,v)$ 有连续偏导数，$z=z(x,y)$ 由 $F\left(x+\dfrac{z}{y},y+\dfrac{z}{x}\right)=0$ 确定，证明 $x\dfrac{\partial z}{\partial x}+y\dfrac{\partial z}{\partial y}=z-xy$.

证：$F_x=F_u-\dfrac{z}{x^2}F_v$，$F_y=F_v-\dfrac{z}{y^2}F_u$，$F_z=\dfrac{F_u}{y}+\dfrac{F_v}{x}$.

故 $x\dfrac{\partial z}{\partial x}+y\dfrac{\partial z}{\partial y}=-\dfrac{xF_x+yF_y}{F_z}=\dfrac{xF_u-\frac{z}{x}F_v+yF_v-\frac{z}{y}F_u}{-(F_u/y+F_v/x)}=\dfrac{(xy-z)\left(\frac{F_u}{y}+\frac{F_v}{x}\right)}{-(F_u/y+F_v/x)}=z-xy$.

10.设 $F(u)$ 有连续导数，函数 $z=z(x,y)$ 由方程 $ax+by+cz=F(x^2+y^2+z^2)$ 所确定.证明：

$$(cy-bz)\frac{\partial z}{\partial x}+(az-cx)\frac{\partial z}{\partial y}=bx-ay.$$

证：设$\varphi(x,y,z)=ax+by+cz-F(x^2+y^2+z^2)$，则有

$$\varphi_x=a-2xF',\ \varphi_y=b-2yF',\ \varphi_z=c-2zF'.$$

于是，左边$=(bz-cy)\dfrac{a-2xF'}{c-2zF'}+(cx-az)\dfrac{b-2yF'}{c-2zF'}\overset{\text{整理}}{=}bx-ay.$

11.设$z^3-3xyz=a^3$，求$\dfrac{\partial^2 z}{\partial x\partial y}$.

解：设$F(x,y,z)=z^3-3xyz-a^3$，则$F_x=-3yz$，$F_y=-3xz$，$F_z=3(z^2-xy)$.

于是，$\dfrac{\partial z}{\partial x}=-\dfrac{F_x}{F_z}=\dfrac{yz}{z^2-xy}$(记为$f(y,z)$)，$\dfrac{\partial z}{\partial y}=\dfrac{xz}{z^2-xy}$.

故$\dfrac{\partial^2 z}{\partial x\partial y}=\dfrac{\partial f}{\partial y}=f_y+f_z\dfrac{\partial z}{\partial y}=\dfrac{z^3}{(z^2-xy)^2}-\dfrac{xyz(xy+z^2)}{(z^2-xy)^3}=\dfrac{z(z^4-2xyz^2-x^2y^2)}{(z^2-xy)^3}$.

12.设函数$z=z(x,y)$由方程$xy+yz+zx=1$所确定，求$\dfrac{\partial^2 z}{\partial x^2},\dfrac{\partial^2 z}{\partial y^2},\dfrac{\partial^2 z}{\partial x\partial y}$.

解：设$F(x,y,z)=xy+yz+zx-1$，则$F_x=y+z$，$F_y=x+z$，$F_z=x+y$.

于是，$\dfrac{\partial z}{\partial x}=-\dfrac{F_x}{F_z}=-\dfrac{y+z}{x+y}\overset{\triangle}{=}f(x,z)$，$\dfrac{\partial^2 z}{\partial x^2}=f_1+f_2\dfrac{\partial z}{\partial x}=\dfrac{2(y+z)}{(x+y)^2}$.

由于原方程对x,y具有轮换对称性，故$\dfrac{\partial z}{\partial y}=-\dfrac{x+z}{x+y}$，$\dfrac{\partial^2 z}{\partial y^2}=\dfrac{2(x+z)}{(x+y)^2}$.

记$\dfrac{\partial z}{\partial x}=g(y,z)$，则$\dfrac{\partial^2 z}{\partial x\partial y}=\dfrac{\partial g}{\partial y}=g_1+g_2\dfrac{\partial z}{\partial y}=\dfrac{2z}{(x+y)^2}$.

注：“轮换对称性”一词已在习题7.2的解答中提及.

13.求由下列方程组所确定的函数的导数或偏导数：

(1)设$\begin{cases}z=x^2+y^2\\x^2+2y^2+3z^2=20\end{cases}$，求$\dfrac{\mathrm{d}y}{\mathrm{d}x},\dfrac{\mathrm{d}z}{\mathrm{d}x}$;

(2)设$\begin{cases}x+y+z=0\\x^2+y^2+z^2=1\end{cases}$，求$\dfrac{\mathrm{d}x}{\mathrm{d}z},\dfrac{\mathrm{d}y}{\mathrm{d}z}$;

(3)设$\begin{cases}x=e^u+u\sin v\\y=e^u-u\cos v\end{cases}$，求$\dfrac{\partial u}{\partial x},\dfrac{\partial u}{\partial y},\dfrac{\partial v}{\partial x},\dfrac{\partial v}{\partial y}$.

解(1)两边对x求导，有$\begin{cases}z'=2x+2yy'\\2x+4yy'+6zz'=0\end{cases}$，解得$\dfrac{\mathrm{d}y}{\mathrm{d}x}=-\dfrac{x(6z+1)}{2y(3z+1)}$，$\dfrac{\mathrm{d}z}{\mathrm{d}x}=\dfrac{x}{3z+1}$.

(2)两边对z求导，有$\begin{cases}x'+y'+1=0\\2xx'+2yy'+2z=0\end{cases}$，解得$\dfrac{\mathrm{d}x}{\mathrm{d}z}=\dfrac{y-z}{x-y}$，$\dfrac{\mathrm{d}y}{\mathrm{d}z}=\dfrac{z-x}{x-y}$.

(3)两边对x求导，有$\begin{cases}1=(e^u+\sin v)u_x+u\cos v\cdot v_x\\0=(e^u-\cos v)u_x+u\sin v\cdot v_x\end{cases}$，解得$\dfrac{\partial u}{\partial x}=\dfrac{\sin v}{e^u(\sin v-\cos v)+1}$，$\dfrac{\partial v}{\partial x}=$

$\dfrac{\cos v-e^u}{ue^u(\sin v-vosv)+u}$.同理，$\dfrac{\partial u}{\partial y}=\dfrac{-\cos v}{e^u(\sin v-\cos v)+1}$，$\dfrac{\partial v}{\partial y}=\dfrac{\sin v+e^u}{ue^u(\sin v-\cos v)+u}$.

14.设$y=f(x,t)$，而t是由$F(x,y,t)=0$确定的x,y的函数，其中f,F都具有一阶连续偏导数.试证：$\dfrac{\mathrm{d}y}{\mathrm{d}x}=$

$\dfrac{f_xF_t-f_tF_x}{f_tF_y+F_t}$.

证：将$\begin{cases}y=f(x,t)\\F(x,y,t)=0\end{cases}$两边对$x$求导，有$\begin{cases}\dfrac{\mathrm{d}y}{\mathrm{d}x}=f_x+f_t\dfrac{\partial t}{\partial x}\\F_x+F_y\dfrac{\mathrm{d}y}{\mathrm{d}x}+F_t\dfrac{\partial t}{\partial x}=0\end{cases}$，解得$\dfrac{\mathrm{d}y}{\mathrm{d}x}=\dfrac{f_xF_t-f_tF_x}{f_tF_y+F_t}$.

15.设$u=f(x,y)$，且$g(x,y,z)=0,h(x,z)=0$，其中各函数都具有连续偏导数，且$g_yh_z\neq0$，求$\dfrac{\mathrm{d}u}{\mathrm{d}x}$.

解：将$\begin{cases}g(x,y,z)=0\\h(x,z)=0\end{cases}$两边对$x$求导，有$\begin{cases}g_x+g_yy_x+g_zz_x=0\\h_x+h_zz_x=0\end{cases}$，解得$y_x=\dfrac{g_zh_x-g_xh_z}{g_yh_z}$.

因此，$\dfrac{\mathrm{d}u}{\mathrm{d}x}=f_x+f_yy_x=f_x+f_y\cdot\dfrac{g_zh_x-g_xh_z}{g_yh_z}$.

16.设函数$f(x,y,z)=x^2yz^3$，其中函数$z=z(x,y)$由方程$x^2+y^2+z^2-3xyz=0$所确定，求$f_x(1,1,1)$.

解：命$F(x,y,z)=x^2+y^2+z^2-3xyz$，则$F_x=2x-3yz$，$F_z=2z-3xy$，$\left.\dfrac{\partial z}{\partial x}\right|_{(1,1)}=-1$.

故$f_x(1,1,1)=\left(2xyz^3+3x^2yz^2\dfrac{\partial z}{\partial x}\right)\Big|_{(1,1,1)}=-1$.

总结：自本节与前面几节的练习，我们很容易看出：求解偏导数的方法并不是单一的！在反复的练习中不难发现有以下几种常见方法：

1.固定其他自变量，然后对须偏导的变量求导；

2.利用全微分公式，可一次性得出所有一阶偏导数；

3.利用隐函数定理.

6.多元函数微分学的几何应用

1.求曲线$x=\dfrac{t}{1+t},y=\dfrac{1+t}{t},z=t^2$在对应$t=1$的切线与法平面方程.

解：易知切线的方向向量的导数为$\left(\dfrac{1}{(1+t)^2},-\dfrac{1}{t^2},2t\right)$代入$t=1$有切线的方向向量为$\left(\dfrac{1}{4},-1,2\right)$，且曲线在$t=1$点的坐标为$\left(\dfrac{1}{2},2,1\right)$，则切线为$\dfrac{x-\frac{1}{2}}{\frac{1}{4}}=\dfrac{y-2}{2}=\dfrac{z-1}{1}$.

故法平面方程为$\dfrac{1}{4}\left(x-\dfrac{1}{2}\right)+2(y-2)+(z-1)=0$.

2.证明螺旋线$x=a\cos t,y=a\sin t,z=bt$上任意一点的切线与z轴成定角.

证：任一点的切线的方向向量为$(-a\sin t,a\cos t,b)$，其与z轴的方向余弦为$\dfrac{b}{\sqrt{b^2+(a\sin t)^2+(a\cos t)^2}}$

$=\dfrac{b}{\sqrt{b^2+a^2}}=$常数，证毕.

3.求曲线$x=t,y=t^2,z=t^3$上的点，使该点处的切线平行于平面$x+2y+z=4$.

解：曲线$x=t,y=t^2,z=t^3$的切线的方向向量为$(1,2t,3t^2)$.

由题意有$1+2t*2+3t^2=0$，解得$t_1=-1$，$t_2=-\frac{1}{3}$.

故点为$(-1,1,-1)$和$\left(-\frac{1}{3},\frac{1}{9},-\frac{1}{27}\right)$.

4.求曲面$e^z-z+xy=3$在点$(2,1,0)$处的切平面与法线方程.

解：曲面$e^z-z+xy=3$在点$(2,1,0)$处的切平面的法向量为(y,x,e^z-1).

代入点有$(1,2,0)$，则切平面方程为$(x-2)+2(y-1)=0$.

因此，法线方程为$\frac{x-2}{1}=\frac{y-1}{2}=\frac{z-0}{0}$.

5.求椭球面$x^2+2y^2+z^2=1$上平行于平面$x-y+2z=0$的切平面方程.

解：椭球面$x^2+2y^2+z^2=1$的切平面的法向量为$(2x,4y,2z)$.

因为与平面$x-y+2z=0$平行，则有$\frac{2x}{1}=\frac{4y}{-1}=\frac{2z}{2}$.

代入$x^2+2y^2+z^2=1$，解得$x=\mp\frac{2}{\sqrt{22}}$,$y=\pm\frac{1}{\sqrt{22}}$,$z=\mp\frac{4}{\sqrt{22}}$.

故切平面方程为$\left(x\mp\frac{2}{\sqrt{22}}\right)-\left(y\pm\frac{1}{\sqrt{22}}\right)+2\left(z\mp\frac{4}{\sqrt{22}}\right)=0$.

6.在曲面$z=xy$上求一点，使该点处的法线垂直于平面$x+3y+z+9=0$.

解：曲面$z=xy$的法线的方向向量为$(y,x,-1)$.

由题意有$\frac{y}{1}=\frac{x}{3}=\frac{-1}{1}$，可得$x=-3,y=-1,z=3$.

故法线为$\frac{x+3}{1}=\frac{y+1}{3}=\frac{z-3}{1}$.

7.求椭球面$\frac{x^2}{a^2}+\frac{y^2}{b^2}+\frac{z^2}{c^2}=1$上在第一卦限上的点，使该点处的法向量的三个方向角相等.

解：因为法向量的三个方向角相等且法向量为$\left(\frac{2x}{a^2},\frac{2y}{b^2},\frac{2z}{c^2}\right)$，则有$\frac{x}{a^2}=\frac{y}{b^2}=\frac{z}{c^2}$.

代入可求得$x=\frac{a}{\sqrt{3}}$,$y=\frac{b}{\sqrt{3}}$,$z=\frac{c}{\sqrt{3}}$.

8.证明曲面$xyz=a^3(a>0)$上任意一点处的切平面与三个坐标面围成的四面体体积为$\frac{9a^3}{2}$.

证：曲面$xyz=a^3(a>0)$上任意一点处的切平面的法向量为(yz,xz,xy).

设点(X,Y,Z)则切平面为$YZ(x-X)+XZ(y-Y)+XY(z-Z)=0$，则与$x\ y\ z$轴的截距依次为$3X$，$3Y,3Z$，故围成的四面体体积为$\frac{1}{6}\cdot3X\cdot3Y\cdot3Z=\frac{9a^3}{2}$.

9.证明曲面$\sqrt{x}+\sqrt{y}+\sqrt{z}=\sqrt{a}$上任意一点处的切平面在各坐标轴上的截距之和等于$a$.

证：曲面$\sqrt{x}+\sqrt{y}+\sqrt{z}=\sqrt{a}$上任意一点处的切平面的法向量为$\left(\frac{1}{2\sqrt{x}},\frac{1}{2\sqrt{y}},\frac{1}{2\sqrt{z}}\right)$.

设点(X,Y,Z)，则切平面为$\frac{1}{2\sqrt{X}}(x-X)+\frac{1}{2\sqrt{Y}}(y-Y)+\frac{1}{2\sqrt{Z}}(z-Z)=0$.

故各坐标轴上的截距之和等于a.

10.设平面$3x+\lambda y-3z+16=0$与椭球面$3x^2+y^2+z^2=16$相切，求λ.

解：设切点为(X,Y,Z)，椭球面$3x^2+y^2+z^2=16$的法向量为$(6X,2Y,2Z)$.

由题意有$\dfrac{6X}{3}=\dfrac{2Y}{\lambda}=\dfrac{2Z}{-3}$，代入椭球面可得$y=\lambda x,z=-3x$.

代入平面和椭球面有$(12+\lambda^2)(x+x^2)=0$，解得$x=0,-1$.

依次代入，经检验去除$x=0$，所以$x=-1$，解之得$\lambda=\pm 2$.

11.求曲线$\begin{cases}xyz=1\\y^2=x\end{cases}$在点$P(1,1,1)$处的切向量的方向余弦.

解：将方程两边对x求导，有$\begin{cases}yz+xy'z+xyz'=0\\2yy'=1\end{cases}$.将$(1,1,1)$代入，解得$y'=z'=\dfrac{1}{2}$.

故切向量为$\left(1,\dfrac{1}{2},\dfrac{1}{2}\right)$，方向余弦为$\dfrac{\sqrt{6}}{3},\dfrac{\sqrt{3}}{3},\dfrac{\sqrt{3}}{3}$.

12.设$f(x,y)$可微分，且对任意的t，有$f(tx,ty)=tf(x,y)$,$P(1,-2,2)$是曲面$z=f(x,y)$上的一点，且$f_x(1,-2)=4$.求曲面在点P处的切平面.

解：由习题7.1解答的拓展，知$xf_x(x,y)+yf_y(x,y)=f(x,y)=z$.

于是，$f_y(1,-2)=1$.即法向量为$(-4,-1,1)$，故切平面为$4x+y-z=4$.

13.求曲面$x=ue^v,y=ve^u,z=u+v$在点$u=0,v=0$处的切平面方程.

解：将$\begin{cases}x=ue^v\\y=ve^u\end{cases}$两边对$x$求导，有$\begin{cases}1=\dfrac{\partial u}{\partial x}e^v+ue^v\dfrac{\partial v}{\partial x}\\0=\dfrac{\partial v}{\partial x}e^u+ve^u\dfrac{\partial u}{\partial x}\end{cases}$.将$(0,0)$代入，有$\begin{cases}\left.\dfrac{\partial u}{\partial x}\right|_{(0,0)}=1\\\left.\dfrac{\partial v}{\partial x}\right|_{(0,0)}=0\end{cases}$.

故$z_x(0,0)=1+0=1$，同理$z_y(0,0)=1+0=1$，即切平面为$z=x+y$.

14.设函数$f(x,y)$在点$(0,0)$附近有定义，且$f_x(0,0)=3,f_y(0,0)=1$，则

$A.\ \mathrm{d}z|_{(0,0)}=3\mathrm{d}x+\mathrm{d}y$;

$B.$曲面$z=f(x,y)$在点$(0,0,f(0,0))$的法向量为$(3,1,1)$;

$C.$曲线$\begin{cases}z=f(x,y)\\y=0\end{cases}$在点$(0,0,f(0,0))$的切向量为$(1,0,3)$;

$D.$曲线$\begin{cases}z=f(x,y)\\y=0\end{cases}$在点$(0,0,f(0,0))$的切向量为$(3,0,1)$.

解(1)没有说偏导数连续，故不确定是否可微分，A不选.

(2)曲面$z=f(x,y)$在点$(0,0,f(0,0))$的法向量为$(3,1,-1)$，B不选.

(3)曲线$\begin{cases}z=f(x,y)\\y=0\end{cases}$在$(0,0,f(0,0))$的切向量是$(1,0,3)$，选$C$.

7. 方向导数和梯度

1.求函数$z=x^2+y^2$在点$(1,2)$处沿从点$(1,2)$到点$\left(2,2+\sqrt{3}\right)$方向的方向导数.

解：点$(1,2)$到点$\left(2,2+\sqrt{3}\right)$方向的方向向量为$\left(\dfrac{1}{2},\dfrac{\sqrt{3}}{2}\right)$，则函数$z=x^2+y^2$在点$(1,2)$沿该方向的方向导

数为$2\cdot\frac{1}{2}+4\cdot\frac{\sqrt{3}}{2}=1+2\sqrt{3}$.

2.求函数$z=1-\left(\frac{x^2}{a^2}+\frac{y^2}{b^2}\right)$在点$\left(\frac{a}{\sqrt{2}},\frac{b}{\sqrt{2}}\right)$处沿曲线$\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$在该点的内法线方向的方向导数.

解：点$\left(\frac{a}{\sqrt{2}},\frac{b}{\sqrt{2}}\right)$沿曲线$\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$在该点的内法线方向的向量为$\left(\frac{-\frac{1}{a}}{\sqrt{\frac{1}{a^2}+\frac{1}{b^2}}},\frac{-\frac{1}{b}}{\sqrt{\frac{1}{a^2}+\frac{1}{b^2}}}\right)$，则函数$z=1-\left(\frac{x^2}{a^2}+\frac{y^2}{b^2}\right)$在点$\left(\frac{a}{\sqrt{2}},\frac{b}{\sqrt{2}}\right)$处的方向导数为$\frac{\frac{\sqrt{2}}{a}\cdot\frac{1}{a}}{\sqrt{\frac{1}{a^2}+\frac{1}{b^2}}}+\frac{\frac{\sqrt{2}}{b}\cdot\frac{1}{b}}{\sqrt{\frac{1}{a^2}+\frac{1}{b^2}}}=\sqrt{2}\sqrt{\frac{1}{a^2}+\frac{1}{b^2}}$.

3.求函数$u=xy^2+z^3-xyz$在点$(1,1,2)$处沿方向角为$\alpha=\frac{\pi}{3},\beta=\frac{\pi}{4},\gamma=\frac{\pi}{3}$的方向的方向导数.

解：$u=xy^2+z^3-xyz$在点$(1,1,2)$处沿方向角为$\alpha=\frac{\pi}{3},\beta=\frac{\pi}{4},\gamma=\frac{\pi}{3}$的方向的方向导数为

$(y^2-yz)\cos\alpha+(2xy-xz)\cos\beta+(3z^2-xy)\cos\gamma=-1\cdot\frac{1}{2}+0+11\cdot\frac{1}{2}=5$.

4.求函数$u=xy^2z$在点$(1,-1,1)$沿什么方向的方向导数最大？并求此方向导数的值.

解：易知，函数$u=xy^2z$在点$(1,-1,1)$沿该点的梯度方向的方向导数最大，则方向导数的最大值为

$y^2z\cdot\frac{1}{\sqrt{3}}+2xyz\cdot\frac{-1}{\sqrt{3}}+xy^2\cdot\frac{1}{\sqrt{3}}=\frac{1}{\sqrt{3}}+\frac{2}{\sqrt{3}}+\frac{1}{\sqrt{3}}=\frac{4}{\sqrt{3}}$.

5.求函数$u=x^2+y^2+z^2$在曲线$x=t,y=t^2,z=t^3$上点$(1,1,1)$处，沿曲线在该点的切线正方向(对应于t增大的方向)的方向导数.

解：曲线$x=t,y=t^2,z=t^3$上点$(1,1,1)$处，沿曲线在该点的切线正方向的方向向量为$(1,2,3)$，则函数$u=x^2+y^2+z^2$在该点的方向导数为$\frac{1}{\sqrt{14}}(2\cdot1+2\cdot2+2\cdot3)=\frac{12}{\sqrt{14}}$.

6.求函数$u=x+y+z$在球面$x^2+y^2+z^2=1$上点$M_0(x_0,y_0,z_0)$处，沿球面在该点的外法线方向的方向导数.

解:球面$x^2+y^2+z^2=1$上点$M_0(x_0,y_0,z_0)$处，沿球面在该点的外法线方向的方向向量为(x_0,y_0,z_0)，则函数$u=x+y+z$在该方向的方向导数为$\frac{x_0+y_0+z_0}{\sqrt{x_0^2+y_0^2+z_0^2}}=x_0+y_0+z_0$.

7.有一山坡，其高度近似为$z=5-x^2-2y^2$，一登山者在坡上$\left(-\frac{3}{2},-1,\frac{3}{4}\right)$处，欲由最陡的道路向上攀登，试问他应该向什么方向行进？

解：由题意可知，应该向函数在该点的梯度方向行进，函数$z=5-x^2-2y^2$在点$\left(-\frac{3}{2},-1,\frac{3}{4}\right)$的梯度为$(2x,4y,1)$.代入点，即向$(-3,-4,1)$行进.

8.一块金属板各点处的温度为$T=xy(1-x)(1-y)\quad(0\le x\le1,0\le y\le1)$，在点$\left(\frac{1}{4},\frac{1}{3}\right)$有一小虫，它为了一条温度下降最快的路线逃离该板，开始时应按什么方向爬行？

解：由题意有应该沿在该点的梯度方向行进，$T=xy(1-x)(1-y)\quad(0\le x\le1,0\le y\le1)$在该点的梯度方

向的方向向量为$((y-y^2)(1-2x),(x-x^2)(1-2y))$.代入数据为$\left(\frac{1}{9},\frac{1}{16}\right)$，即沿该方向爬行.

9.设函数$f(x,y)=\begin{cases}\frac{x^3}{y} & y\neq 0\\ 0 & y=0\end{cases}$，证明函数在点$(0,0)$处不连续，但在该点的任意方向的方向导数存在.

证：因为$\lim\limits_{x\to 0,y\to 0} f(x,y)\overset{y=kx^3}{=}\frac{x^3}{kx^3}=\frac{1}{k}$不为常数，即在该点不连续，在该点的任意方向的方向向量为$(\cos a,\cos b)$.其中$\cos^2 a+\cos^2 b=1$，所以方向导数为$0\cdot\cos a+0\cdot\cos b=0$，证毕.

8.多元函数的极值及其求法

1.求下列函数的极值

(1)$f(x,y)=x^3+3xy^2-15x-12y$ (2)$f(x,y)=(6x-x^2)(4y-y^2)$

(3)$f(x,y)=e^{2x}(x+y^2+2y)$ (4)$f(x,y)=\frac{8}{x}+\frac{x}{y}+y(x,y>0)$ (5)$f(x,y)=1-(x^2+y^2)^{\frac{2}{3}}$

解(1)令$f_x=f_y=0$，有$\begin{cases}x=1\\ y=2\end{cases}$,或$\begin{cases}x=-1\\ y=-2\end{cases}$,或$\begin{cases}x=2\\ y=1\end{cases}$,或$\begin{cases}x=-2\\ y=-1\end{cases}$.

满足$f_{xx}f_{yy}-f_{xy}^2=36(x^2-y^2)>0$的只有解$(2,1)$和$(-2,-1)$.

故$f_{极大}(x,y)=f(-2,-1)=28$，$f_{极小}(x,y)=f(2,1)=-28$.

(2)令$\begin{cases}f_x=f_y=0\\ f_{xx}f_{yy}-f_{xy}^2>0\end{cases}$，有$\begin{cases}x=3\\ y=2^{-1}\end{cases}$，于是$f_{极小}(x,y)=f(3,2^{-1})=\frac{63}{4}$，无极大值.

(3)令$f_x=f_y=0$，解得$x=\frac{1}{2},y=-1$，代入$f_{xx}f_{yy}-f_{xy}^2$，计算得$4e>0$.

故$f_{极小}(x,y)=f\left(\frac{1}{2},-1\right)=-\frac{e}{2}$，无极大值.

(4)令$f_x=f_y=0$，有$x=4,y=2$，此时$f_{xx}f_{yy}-f_{xy}^2>0$，故$f_{极小}(x,y)=6$，无极大值.

(5)显然，无极小值，极大值为$f(0,0)=1$.

2.求由方程$x^2+y^2+z^2-2x+2y-4z-10=0$所确定的函数$z=z(x,y)$的极值.

解：易见，$(x-1)^2+(y+1)^2+(z-2)^2=16$.故极大值为$4+2=6$，极小值为$-4+2=-2$.

3.求函数$z=xy$在适合约束条件$x+y=1$下的极大值.

解：显然，$z=xy\leqslant\frac{1}{4}(x+y)^2=\frac{1}{4}$，故极大值为$\frac{1}{4}$.

4.求函数$z=xy(4-x-y)$在直线$x=1,y=0,x+y=6$围成的闭区域上的最值.

解：在$x=1$的最值是$\frac{9}{4},-10$;在$y=0$的最值是0;在$x+y=6$的最值是$0,-18$.

令$\begin{cases}f_x=f_y=0\\ f_{xx}f_{yy}-f_{xy}^2>0\end{cases}$，有$x=y=\frac{4}{3}$，极大值为$\frac{64}{27}$.

综上，最大值为$\frac{64}{27}$，最小值为-18.

5.从斜边长为l的一切直角三角形中，求有最大周长的直角三角形.

解：设直角边长分别为x,y，则$x^2+y^2=l^2$.

故周长为$x+y+l\leqslant\sqrt{2}\sqrt{x^2+y^2}+l=\left(\sqrt{2}+1\right)l$.

从取等条件知，这样的三角形边长分别为$\frac{l}{\sqrt{2}},\frac{l}{\sqrt{2}},l$.

6.求抛物线$y=x^2$到直线$x-y-2=0$之间的最短距离.

解：令$y'=2x=1$，有$x=\frac{1}{2}$.故最短距离为$d=\frac{\left|\frac{1}{2}-\frac{1}{4}-2\right|}{\sqrt{2}}=\frac{7\sqrt{2}}{8}$.

7.要造一个容积为k的长方体无盖水池，应如何选择水池的尺寸，方可使其表面积最小？

解：设长宽分别为x,y，则表面积$S=xy+2yz+2xz=xy+\frac{2k}{x}+\frac{2k}{y}\geqslant\sqrt[3]{4k^2}$.

由取等条件知，选取尺寸为$\sqrt[3]{2k},\sqrt[3]{2k},\sqrt[3]{\frac{k}{4}}$.

8.在xOy平面上求一点，使它到$x=0,y=0,x+2y-16=0$的距离平方和最小.

解：设点(x,y)，则距离平方和$z=x^2+y^2+\frac{1}{5}(x+2y-16)^2\geqslant25.6$.

由Cauchy不等式取等条件，点是$\left(\frac{8}{5},\frac{16}{5}\right)$.

9.将周长为$2p$的矩形绕它的一边旋转而成一个圆柱体，问矩形的边长各为多少时，才能使圆柱体积最大？

解：设一边长为x，则体积$V=\frac{\pi}{4}x^2(p-x)\leqslant\frac{\pi}{27}p^3$，当且仅当$x=\frac{2p}{3}$时，等号成立.

故边长为$\frac{2p}{3},\frac{p}{3}$时，圆柱体积最大.

10.做一个长方体的箱子，其容积为$\frac{9}{2}\mathrm{m}^3$，箱子的盖及侧面的造价为8元$/\mathrm{m}^3$，箱底的造价为1元$/\mathrm{m}^3$.试求造价最低的箱子的尺寸.

解：设长宽分别为x,y，则表面积$S=16xy+16xz+9yz$

$=16xy+\frac{72}{y}+\frac{81}{2x}\geqslant36$，当且仅当$x=\frac{9}{8},y=2$时取等.

故尺寸为$1.125\mathrm{m},2\mathrm{m},2\mathrm{m}$.

11.在椭球面$x^2+y^2+\frac{z^2}{4}=1$的第一卦限部分求一点，使该点处的切平面在坐标轴的截距平方和最小.

解：设$P(x_0,y_0,z_0)$，则P点的切平面为$4x_0x+4y_0y+z_0z=4$，截距平方和$s=x_0^{-2}+y_0^{-2}+16z_0^{-2}$.

由Cauchy不等式取等条件，s最小，当且仅当切点为$\left(\frac{1}{2},\frac{1}{2},\sqrt{2}\right)$.

12.在球面$x^2+y^2+z^2=1$上求一点，使它到点$P(1,2,3)$的距离最远.

解：由几何关系，球面上的点$A\left(\frac{1}{\sqrt{14}},\frac{2}{\sqrt{14}},\frac{3}{\sqrt{14}}\right)$在$OP$上.故$A'\left(-\frac{1}{\sqrt{14}},-\frac{2}{\sqrt{14}},-\frac{3}{\sqrt{14}}\right)$为所求.

13.用极值方法求点$(0,1,1)$到直线$\begin{cases}x+2z=7\\y=-2\end{cases}$的距离.

解：设直线上的点$P(7-2z,-2,z)$，则距离平方$s=5z^2-30z+59\geqslant 14$，故距离为$\sqrt{14}$.

14.求椭圆抛物面$z=x^2+y^2$到平面$x+y-z=1$的最短距离.

解：设$P(x,y,x^2+y^2)$，则距离平方$s=\dfrac{1}{3}(x+y-x^2-y^2-1)^2\geqslant\dfrac{1}{12}$.故最短距离是$\dfrac{\sqrt{3}}{6}$.

15.求函数$f(x,y)=(x-1)^2+(y-2)^2+1$在区域$D=\{(x,y)\mid x^2+y^2\leqslant 20\}$的最值.

解：显然，点$A(1,2)\in D$，且$f(x,y)-1$表示D上的点到A的距离平方s.

由几何关系，$s_{\max}=2\sqrt{5}+\sqrt{5}=3\sqrt{5}$.故最大值为$3\sqrt{5}$，最小值为$1$.

16.在椭圆$x^2+4y^2=4$上求一点，使其到直线$2x+3y=6$的距离最近.

解：椭圆上点(x,y)处的切向量为$(x,4y)$，与$(2,3)$正交，即$x+6y=0$.代入椭圆方程，得$\begin{cases}x=\dfrac{8}{5}\\y=\dfrac{3}{5}\end{cases}$.

17.设函数$f(x,y)$在$(0,0)$某邻域连续，且$\lim\limits_{x\to 0,y\to 0}\dfrac{f(x,y)-xy}{(x^2+y^2)^2}=1$，则（　）

$A.(0,0)$不是$f(x,y)$的极值点 $B.(0,0)$是$f(x,y)$的极大值点 $C.(0,0)$是$f(x,y)$的极小值点 $D.$不能判定

解：$(x,y)\to(0,0)$时，$z=f(x,y)=xy+(x^2+y^2)^2+\varepsilon\rho^4\left(|\varepsilon|<\dfrac{1}{4}\right)$.

令$y=x$，则$z=x^2+(4+4\varepsilon)x^4>0$;令$y=-x$，当$x^2<\dfrac{1}{4}$时，$z=(4+4\varepsilon)x^4-x^2<0$.

故$(0,0)$不是极值点，选A.

总习题

1.设函数$f(x,y)$在点$(0,0)$的某邻域内有定义，且$f_x(0,0)=3,f_y(0,0)=-1$,则有（　）.

$(A)\ dz|_{(0,0)}=3dx-dy.$

(B)曲面$z=f(x,y)$在点$(0,0,f(0,0))$的一个法向量为$(3,-1,1)$.

(C)曲线$\begin{cases}z=f(x,y)\\y=0\end{cases}$在点$(0,0,f(0,0))$的一个切向量为$(1,0,3)$.

(D)曲线$\begin{cases}z=f(x,y)\\y=0\end{cases}$在点$(0,0,f(0,0))$的一个切向量为$(3,0,1)$.

解：本题完全类似于6的题14，解析见6.

2.如果一元函数$f(x_0,y)$在点y_0连续，$f(x,y_0)$在点x_0连续,那么二元函数$f(x,y)$在(x_0,y_0)处是否必连续？

解：设$f(x,y)=\dfrac{xy}{x+y}$，在$(x_0,y_0)=(0,0)$处延拓为0.

3.设函数$z=f(u,v,w)$有连续偏导数,而$u=\eta-\xi,v=\xi-\zeta,w=\zeta-\eta$,求$\dfrac{\partial z}{\partial\xi},\dfrac{\partial z}{\partial\eta},\dfrac{\partial z}{\partial\zeta}$.

解：$\dfrac{\partial z}{\partial\xi}=f_u(-1)+f_v\cdot 1=f_v-f_u$.同理$\dfrac{\partial z}{\partial\eta}=f_u-f_w$，$\dfrac{\partial z}{\partial\zeta}=f_w-f_v$.

4.求函数$f(x,y)=x^2-xy+y^2$在点$P(1,1)$沿与x轴正方向夹角为α的方向的方向导数，并分别确定角α,使这方向导数有(1)最大值,(2)最小值,(3)等于零.

解：$f_x(1,1)=f_y(1,1)=1$，且方向向量$\boldsymbol{l}=(\cos\alpha,\sin\alpha)$.于是，方向导数为$\sqrt{2}\sin\left(\alpha+\dfrac{\pi}{4}\right)$.

(1)$\alpha=\dfrac{\pi}{4}$时，取最大值；(2)$\alpha=\dfrac{5\pi}{4}$时，取最小值；(3)$\alpha=\dfrac{3\pi}{4}$或$\dfrac{7\pi}{4}$时，为零.

5.证明曲面$F(y-az,x-bz)=0$上任一点处的切平面与一常向量平行,其中a,b均为非零常数.

证：注意到$F_x=F_2$，$F_y=F_1$，$F_z=-aF_1-bF_2$，则法向量$\boldsymbol{n}$与$\boldsymbol{l}=(b,a,1)$正交，证毕.

6.证明曲面$F\left(\dfrac{y-b}{x-a},\dfrac{z-c}{x-a}\right)=0$上任一点处的切平面通过某定点.

证：设$P(x_0,y_0,z_0)$，则P点的法向量为$\boldsymbol{n}=\left(\dfrac{(y_0-b)F_1+(z_0-c)F_2}{-(x_0-a)^2},\dfrac{F_1}{x_0-a},\dfrac{F_2}{x_0-a}\right)$.

$\therefore P$点的切平面是$\dfrac{(y_0-b)F_1+(z_0-c)F_2}{-(x_0-a)^2}(x-x_0)+\dfrac{F_1}{x_0-a}(y-y_0)+\dfrac{F_2}{x_0-a}(z-z_0)=0$.

令(a,b,c)代入上式，原式恒成立.故过定点(a,b,c).

7.求函数$f(x,y)=(x^2+y^2)e^{-(x^2+y^2)}$的极值.

解：易知当x与y不同时为0时均有$f(x,y)\geqslant f(0,0)=0$，所以原点为极小值点.

令$x^2+y^2=r^2$，则$f(x,y)=g(r^2)=r^2e^{-r^2}\leqslant\dfrac{1}{e}$，故极大值为$\dfrac{1}{e}$.

8.求正数λ，使曲面$xyz=\lambda$及$\dfrac{x^2}{a^2}+\dfrac{y^2}{b^2}+\dfrac{z^2}{c^2}=1$在第一卦限某点相切，并求切点坐标.

解：将$z=\dfrac{\lambda}{xy}$代入椭球方程，有$1=\dfrac{x^2}{a^2}+\dfrac{y^2}{b^2}+\dfrac{\lambda^2}{c^2}\cdot\dfrac{1}{x^2y^2}\geqslant 3\sqrt[3]{\dfrac{\lambda^2}{a^2b^2c^2}}$.解得$\lambda\leqslant\dfrac{abc}{3\sqrt{3}}$.

由于相切，因此λ必须达到最大值，即$\lambda=\dfrac{abc}{3\sqrt{3}}$.根据取等条件，切点为$\left(\dfrac{a}{\sqrt{3}},\dfrac{b}{\sqrt{3}},\dfrac{c}{\sqrt{3}}\right)$.

9.求曲面$x=ue^v,y=ve^u,z=u+v$在对应$u=0,v=0$的切平面方程.

解：本题6的题13完全相同，见6的解答.

10.求平面$\dfrac{x}{3}+\dfrac{y}{4}+\dfrac{z}{5}=1$和柱面$x^2+y^2=1$的交线上与$xoy$面距离最短的点.

解：利用Lagrange乘数法解得当$(x,y)=\left(\dfrac{4}{5},\dfrac{3}{5}\right)$时取最短的点，代入得此时$z=\dfrac{35}{12}$.

11.设函数$f(x,y,z)$满足$f(tx,ty,tz)=t^kf(x,y,z)$,其中f有连续偏导数,证明$x\dfrac{\partial f}{\partial x}+y\dfrac{\partial f}{\partial y}+z\dfrac{\partial f}{\partial z}=kf$.

证：见1的拓展1.

12.设$\begin{cases}u=f(ux,v+y)\\v=g(u-x,v^2y)\end{cases}$其中$f,g$有一阶连续偏导数,求$\dfrac{\partial u}{\partial x},\dfrac{\partial v}{\partial x}$.

解：将原式两边对x求导，有$\begin{cases}u_x=f_1(u+xu_x)+f_2v_x\\v_x=g_1(u_x-1)+2g_2vv_x\end{cases}$.

解得$u_x=\dfrac{-uf_1(2yvg_2-1)-f_2g_1}{(xf_1-1)(2yvg_2-1)-f_2g_1}$，$v_x=\dfrac{g_1(xf_1+uf_1-1)}{(xf_1-1)(2yvg_2-1)-f_2g_1}$.

13.第一卦限内求椭球面$\dfrac{x^2}{a^2}+\dfrac{y^2}{b^2}+\dfrac{z^2}{c^2}=1$的切平面,使该切平面与三坐标面所围成的四面体体积为最小,并求此最小体积.

解：点(x_0,y_0,z_0)处的切平面方程为$\dfrac{x_0}{a^2}x+\dfrac{y_0}{b^2}y+\dfrac{z_0}{c^2}z=1$，围成的体积$V=\dfrac{abc}{6\sqrt{\frac{x_0^2}{a^2}\frac{y_0^2}{b^2}\frac{z_0^2}{c^2}}}\geqslant\dfrac{\sqrt{3}abc}{2}$.

由取等条件，切平面为$\dfrac{x}{a}+\dfrac{y}{b}+\dfrac{z}{c}=\sqrt{3}$.

14.在半径为R的圆的内接三角形中，求出其面积最大者.

解：设三角形三定点对应的圆心角分别为α,β,γ.

于是，面积$S=\dfrac{R^2}{2}(\sin\alpha+\sin\beta+\sin\gamma)\leqslant\dfrac{3}{2}R^2\sin\dfrac{\alpha+\beta+\gamma}{3}=\dfrac{3\sqrt{3}}{4}R^2$.

上述不等式是由于$\sin x$在$(0,\pi)$是下凸的，且可知：面积最大者是正三角形.

15.将正数a分成三个正数x,y,z之和，使xyz^3最大，并证明$xyz^3\leqslant27\left(\dfrac{x+y+z}{5}\right)^5$.

证：$xyz^3=27x\cdot y\cdot\dfrac{z}{3}\cdot\dfrac{z}{3}\cdot\dfrac{z}{3}\leqslant27\left(\dfrac{x+y+z/3+z/3+z/3}{5}\right)^5=27\left(\dfrac{x+y+z}{5}\right)^5$.

因此，可将a分成$\dfrac{a}{5},\dfrac{a}{5},\dfrac{3a}{5}$之和.

16.求由方程$2x^2+2y^2+z^2+yz-2xy+2x-4y-22=0$所确定的函数$z=z(x,y)$的极值.

解：$\dfrac{\partial z}{\partial x}=\dfrac{2y-4x-2}{2z+y}$，$\dfrac{\partial z}{\partial y}=\dfrac{2x-4y-z+4}{2z+y}$.令之为零，解得$\begin{cases}x=1\\y=3\\z=-6\end{cases}$或$\begin{cases}x=-\dfrac{4}{5}\\y=-\dfrac{3}{5}\\z=\dfrac{24}{5}\end{cases}$.

经检验，-6和$\dfrac{24}{5}$均为$z=z(x,y)$的极值.

17.在圆锥面$Rz=h\sqrt{x^2+y^2}$与$z=h$ $(h>0,R>0)$所围成的圆锥内作一个底面平行于xoy面的长方体，使长方体的体积为最大，求出其体积.

解：设长方体的长宽高分别为a,b,c，则当其内接于圆锥时才可能有最大体积.

注意到$r=R\left(1-\dfrac{c}{h}\right)$，且$a^2+b^2=r^2$.

故体积$V=abc\leqslant\dfrac{r^2}{2}c=R^2h\cdot\dfrac{c}{2h}\left(1-\dfrac{c}{h}\right)\left(1-\dfrac{c}{h}\right)\leqslant R^2h\left(\dfrac{c/2h+1-c/h+1-c/h}{3}\right)^3=\dfrac{8}{27}R^2h$.

验证取等条件后，可得$V_{\max}=\dfrac{8}{27}R^2h$.

18.证明函数$f(x,y)=(1+e^y)\cos x-ye^y$有无穷多个极大值而无极小值.

证：令$f_x=f_y=0$，解得$x=n\pi,y=1-\cos x(n\in\mathbb{Z})$.

经检验，只有$x=2n\pi,y=0(n\in\mathbb{Z})$满足$f_{xx}f_{yy}-f_{xy}^2>0$，故$f(x,y)$只能取极大值.

19.设$f(x,y)=\begin{cases}\dfrac{x^2y^2}{(x^2+y^2)^{\frac{3}{2}}}, & x^2+y^2\neq 0\\ 0, & x^2+y^2=0\end{cases}$.证明$f(x,y)$在点$(0,0)$连续且偏导数存在，但不可微分.

证：注意到$\dfrac{x^2y^2}{(x^2+y^2)^{\frac{3}{2}}}\leqslant\dfrac{(x^2+y^2)^2}{4(x^2+y^2)^{\frac{3}{2}}}=\dfrac{\sqrt{x^2+y^2}}{4}$，即$\lim\limits_{(x,y)\to(0,0)}f(x,y)=f(0,0)=0$.

又$\lim\limits_{x\to 0}\dfrac{f(x,0)-f(0,0)}{x}=\lim\limits_{y\to 0}\dfrac{f(0,y)-f(0,0)}{y}=0$，即$f_x(0,0)=f_y(0,0)=0$.

而$\Delta z=f(\Delta x,\Delta y)\neq o(\rho)$，因此$f(x,y)$在$(0,0)$是不可微的.

20.证明函数$f(x,y)=\begin{cases}x+\dfrac{x^5}{(y-x^2)^2+x^6}, & x^2+y^2\neq 0\\ 0, & x^2+y^2=0\end{cases}$在点$(0,0)$处，沿任意方向的方向导数都存在，但$f(x,y)$在该点不连续.

证：$f_x(0,0)=\lim\limits_{x\to 0}\dfrac{f(x,0)-f(0,0)}{x}=2$，$f_y(0,0)=0$，故沿任意方向的方向导数都存在.

由$f(0,0)=0\neq 1=\lim\limits_{x\to 0,y=x^2+\sqrt{|x^5-x^6|}}f(x,y)$，故$f(x,y)$在$(0,0)$不连续.

21.设$x^2=vw,y^2=uw,z^2=uv$，且$f(x,y,z)=F(u,v,w)$，f可微，F由前三式代入f中得到.试证：$xf_x+yf_y+zf_z=uF_u+vF_v+wF_w$.

证：由前三个等式，解得$u^2=\dfrac{y^2z^2}{x^2}$，两边再对x,y,z分别偏导，相加得$xu_x+yu_y+zu_z=u$.

将$f(x,y,z)=F(u,v,w)$两边对x偏导，有$xf_x=F_uxu_x+F_vxv_x+F_wxw_x$.

于是，$xf_x+yf_y+zf_z=F_u(xu_x+yu_y+zu_z)+F_v(xv_x+yv_y+zv_z)+F_w(xw_x+yw_y+zw_z)$

$=uF_u+vF_v+wF_w$.

22.若任取$t>0$，$f(x,y,z)$满足$f(tx,ty,tz)=t^kf(x,y,z)$.则称函数$f(x,y,z)$为k次齐次函数，证明函数$f(x,y,z)$为k次齐次函数的充要条件是$xf_x(x,y,z)+yf_y(x,y,z)+zf_z(x,y,z)=kf(x,y,z)$.

证：见1的拓展1.

注：这份多元函数练习题是东北大学2017级高等数学下册的教材课后习题，现分享给大家，笔者并将它们全部解出，以希读者能更加熟练地掌握多元函数的基础计算题目.这些题目中笔者有意避开Lagrange乘数法的使用，因为很多题目直接利用不等式会更加简洁，望读者自己加以练习和掌握.

——2019.2.24晚23:47于东北大学